

VF vs VU

Towards a filter version of virtual ultracategories
(based on a join work in progress with Umberto Tarantino)

Gabriel Saadia

IRIF: Ultracategories and Stone Duality Workshop

June 30, 2026



Stockholm
University

Filters, ultrafilters and convergence

Categorifying topological spaces

Toposes categorify locales that are a point free version of topological spaces.

A notion of categorified space should fit in the ? below.

topological spaces	locales
?	toposes

Garner's ionads [Gar12] (generalized by Di Liberti in [DL22b]) categorify the interior comonad. They end up to be quite close to toposes.

Last summer, a new proposal of such a notion has been introduced independently in three different works [Saa25, Ham25, vGMT26] dubbed **virtual ultracategories** (abbreviated **vu-categories**) in the first above-cited work.

Vu-categories categorify topological spaces seen as a collection of points with some **convergence data**.

Convergence

Let σ be a filter on S . Recall that points $(q_s)_{s \in S}$ converge to p along σ if

$$(\forall U \text{ open}) \ p \in U \Rightarrow (\forall s : \sigma) \ q_s \in U$$

which we write $p \preccurlyeq_\sigma q_s$.

The ultraconvergence relation is enough to recover the topology: a subset A is open if and only if for each $p \preccurlyeq_\sigma q_s$ we have $p \in A \Rightarrow (\forall s : \sigma) \ q_s \in A$.

One can check that $\preccurlyeq$ behaves like a partial order

$$p \preccurlyeq_\star p \quad (p \preccurlyeq_\sigma q_s \text{ and } (\forall s : \sigma) \ q_s \preccurlyeq_{\tau_s} r_{s,t}) \Rightarrow p \preccurlyeq_{\sigma \cdot \tau_s} r_{s,t}$$

that extends the usual specialization order

$$p \preccurlyeq_\star q \text{ if and only if } p \leq^{sp} q.$$

Topologies can be seen as a **posetal generalized multicategorical structure**, with arrows of the shape

$$p \preccurlyeq_\sigma q_s$$

and vu-categories categorify this point of view.

The category of filters

For σ a filter on S , the topological space $A_{S,\sigma}$ is given by the set of points $\{a\} \sqcup \{b_s | s \in S\}$ and opens are the subsets U such that

$$a \in U \Rightarrow (\forall s : \sigma) b_s \in U.$$

$A_{S,\sigma}$ represents the formal σ -convergence of $(b_s)_{s \in S}$ points to a , and σ can be seen as the germ of a in $A_{S,\sigma}$.

The category of filters $\mathbb{F}$ can be defined as follows [VJ70].

- Objects are pairs (S, σ) with σ a filter on S .
- A morphism of filters $(T, \tau) \rightarrow (S, \sigma)$ is a partial map $T_0 \rightarrow S$ with τ -large domain, such that the preimage of a σ -large subset is τ -large, and two such maps are identified if they coincide on a τ -large subset.

$\mathbb{F}$ -morphisms $(T, \tau) \rightarrow (S, \sigma)$ can be identified with continuous maps of germs $(A_{T,\tau}, a) \rightarrow (A_{S,\sigma}, a)$ sending (b_s) to the (b_t) .

We denote by $\mathbb{U} \subseteq \mathbb{F}$ the full subcategory of ultrafilters.

Morphisms between filters

A map $f : \tau \twoheadrightarrow \sigma$ is epic if the image of a σ -large set is τ -large, i.e. if $\sigma = f_*(\tau)$.

Epimorphisms in $\mathbb{F}$ are also called **covering**. Every map in $\mathbb{U}$ is a covering.

Note that a map is a covering if and only if it induces an open surjection of germs.

A map $f : \tau \hookrightarrow \sigma$ is monic if it can be represented by an injection $T_0 \hookrightarrow S$. In that case, we say that τ is a **specialization** of σ .

In particular, ultrafilters extending a filter σ are given exactly by subobjects $\lambda \hookrightarrow \sigma$ with λ ultrafilter.

From [Bla77, Moe95], $\mathbb{F}$ has **small coproducts**, is a **regular** category, and **every epimorphism is regular**. A map $f : \tau \rightarrow \sigma$ factorizes as

$$\tau \twoheadrightarrow f_*(\tau) \hookrightarrow \sigma.$$

Reduced product

We identify a set A and the principal filter at A (with only A large).

We have $A^\sigma = \mathbb{F}(\sigma, A)$, an element of A^σ is called a **σ -family** of element of A and is denoted by $(a_s)_{s:\sigma}$.

In the same way, the element of the reduced/ultra-product $\int_{s:\sigma} A_s$ can be seen as the set of sections in $\mathbb{F}$:

$$\begin{array}{ccc} & & \bigsqcup_{s \in S} A_s \\ & \nearrow \text{dashed} & \downarrow \\ \sigma & \longrightarrow & S \end{array}$$

An element of $\int_{s:\sigma} A_s$ is a **dependent σ -family** and denoted by $(a_s \in A_s)_{s:\sigma}$.

Precomposition by $\phi : \tau \rightarrow \sigma$ gives a **reindexing map** $\phi^* : \int_{s:\sigma} A_s \rightarrow \int_{t:\tau} A_{\phi t}$

$$\phi^*(a_s)_{s:\sigma} = (a_{\phi t})_{t:\tau}.$$

Tensor

The tensor product on the category of filters is usually denoted by $\sigma \otimes \tau$, we use $\sigma \cdot \tau$

The **dependent tensor** of $(\tau_s)_{s:\sigma}$ is a filter $\sigma \cdot \tau_s$ defined on the set $\bigsqcup_{s \in S} T_s$ (the indexed sum of [Gar20]).

The $\sigma \cdot \tau_s$ -large sets are the $\bigsqcup_{s \in S} R_s \subseteq \bigsqcup_{s \in S} T_s$ with σ -all R_s being τ_s -large.

It can be seen as gluing the spaces A_{T, τ_s} with $A_{S, \sigma}$.

Alternatively, it can be seen as the following pullback in $\mathbb{F}$

$$\begin{array}{ccc} \sigma \cdot \tau_s & \hookrightarrow & \bigsqcup_{s \in S} \tau_s \\ \downarrow & \lrcorner & \downarrow \\ \sigma & \hookrightarrow & S \end{array} \quad \text{in particular, } S \cdot \tau_s = \bigsqcup_{s \in S} \tau_s.$$

Vu-categories

Vu-categories

A vu-category is a categorical notion where arrows have the shape

$$p \multimap_{s:\sigma} q_s$$

Such a vu-arrow can either be seen as categorifying $p \preccurlyeq_{s:\sigma} q_s$ or as a morphism $p \rightarrow \int_{s:\sigma} q_s$ into some *virtual* ultraproduct of the $(q_s)_{s:\sigma}$.

Vu-categories 1/3

A vu-category $\mathcal{X}$ is given by

- a class of objects usually called **points**.
- a class of **vu-arrows**

$$u : p \multimap_{s:\sigma} q_s$$

with domain a point p , and codomain a σ -family of points $(q_s)_{s:\sigma}$ for some ultrafilter σ .

We denote by $\mathcal{X}_\sigma(p, q_s)$ the class of vu-arrows of domain p and codomain $(q_s)_{s:\sigma}$.

Vu-categories: composition

Composition of vu-arrows relies on the dependent tensor of ultrafilters.

Vu-categories 2/3: composition

- for each point p , a **unit** $i_p : p \multimap_\star p$
- and for $u : p \multimap_\sigma q_s$ and $(v_s : q_s \multimap_{\tau_s} r_{s,t})_{s:\sigma}$, a **composite**

$$v_s \circ_\sigma u : p \multimap_{\sigma \cdot \tau_s} r_{s,t}$$

With the expected neutrality and associativity axioms

$$u \circ_\star i_p = u \quad i_{q_s} \circ_\sigma u = u$$

$$(w_{s,t} \circ_{\tau_s} v_s) \circ_\sigma u = w_{s,t} \circ_{\sigma \cdot \tau_s} (v_s \circ_\sigma u)$$

Vu-categories: reindexings

More subtle, we need a notion of **reindexing** of vu-arrows:

Vu-categories 3/3: reindexing

- for $u: p \multimap_{\sigma} q_s$ and $\phi: \tau \rightarrow \sigma$, a **reindexing** $\phi^* u: p \multimap_{\tau} q_{\phi t}$

The reindexing operation is functorial

$$\text{id}_{\sigma}^* u = u \quad \psi^*(\phi^* u) = (\phi \circ \psi)^* u$$

and compatible with the composition

$$(\psi_s^* v_s) \circ_{\sigma} u = (\text{id}_{\sigma} \cdot \psi_s)^*(v_s \circ_{\sigma} u) \quad v_{\phi \ell} \circ_{\lambda} (\phi^* u) = (\phi \cdot \text{id}_{\tau_s})^*(v_s \circ_{\sigma} u)$$

This concludes the definition of vu-categories.

We call **vu-posets** the vu-categories with the homsets being subsingletons.

We have a 2-category of vu-categories **vuCat** given by vu-functors and natural transformations between vu-functors.

The reindexing functor

For points p and $(q_s)_{s:\sigma}$ the reindexing induces a **reindexing functor**

$$\begin{aligned}\mathcal{X}_\sigma^{p,q_s} : (\mathbb{U}/\sigma)^{\text{op}} &\rightarrow \text{SET} \\ (\phi : \tau \rightarrow \sigma) &\mapsto \mathcal{X}_\tau(p, q_{\phi t})\end{aligned}$$

Thickening

All map $\tau \rightarrow \sigma$ in $\mathbb{U}$ are epis, hence we call $\phi^* u$ the **thickening** of u along ϕ .

Locally small and small

$\mathcal{X}$ is said *locally small* if these functors are small presheaves, and *small* if moreover its collection of points is small.

$\mathbf{Vu}$ -categories are always assumed locally small.

Merging

Diagonal

In particular, thickening the identity $i_p : p \multimap_\star p$ along the unique map $\sigma \rightarrow \star$ gives a vu-arrow $d_\sigma^p : p \multimap_\sigma p$ called a **diagonal of p** .

The diagonal enables us to **merge** together vu-arrows with common domain.

Merging

The **merging** of a σ -family of vu-arrows $(v_s : p \multimap_{\tau_s} q_{s,t})_{s:\sigma}$ is the composite

$$v_s \circ_\sigma d_\sigma^p : p \multimap_{\sigma \cdot \tau_s} q_{s,t}.$$

The merging induces $\int_{\ell:\lambda} \mathcal{X}_{\tau_\ell}(p, q_{\phi_\ell t}) \rightarrow \mathcal{X}_{\lambda \cdot \tau_s}(p, q_{\phi_\ell t})$ for $(\phi_\ell : \tau_\ell \rightarrow \sigma)_{\ell:\lambda}$ and thus gives a structure of **right ultrafunctor** on the reindexing functor

$$\mathcal{X}_\sigma^{p,q_s} : (\mathbb{U}/\sigma)^{\text{op}} \rightarrow \text{Set},$$

or equivalently an ultrastructure on the category of elements of this presheaf.

Vu-categories: examples and reconstruction theorem for toposes

Example: topological spaces

Every topological space X has a vu-poset structure given by $p \preceq_\sigma q_s$.

The reindexing operation gives (for $\phi : \tau \rightarrow \sigma$)

$$p \preceq_\sigma q_s \Rightarrow p \preceq_\tau q_{\phi t}.$$

Proposition

We can identify topological spaces to the small vu-posets satisfying

$$p \preceq_\sigma q_s \Leftrightarrow p \preceq_\tau q_{\phi t}$$

Example: ultracategories

An ultracategory¹ $\mathcal{M}$ has a vu-structure with vu-arrows $p \multimap_{\sigma} q_s$ given by maps $p \rightarrow \int_{s:\sigma} q_s$.

$$\mathcal{M}_{\sigma}(p, q_s) = \mathcal{M}(p, \int_{s:\sigma} q_s)$$

The reindexing of $p \multimap_{\sigma} q_s$ along $\phi : \tau \rightarrow \sigma$ corresponds to the composition:

$$\begin{array}{ccc} & \int_{t:\tau} q_{\phi t} & \\ \phi^* u \nearrow & \uparrow \phi^* & \\ p & \xrightarrow{u} & \int_{s:\sigma} q_s \end{array}$$

justifying the notation $\phi^* u$.

In particular, we get a vu-category structure on $\mathbf{Set}$, and we can show:

Theorem [Saa25, Ham25, vGMT26]

Let X be a topological space, $\mathbf{Sh}(X) \simeq \mathbf{vuCat}(X, \mathbf{Set})$.

¹ultracategory are always assumed strong

Example: points of a topos

For $\mathcal{E}$ a topos, we have a vu-structure on the points of $\mathcal{E}$ with vu-arrows $p \multimap_{\sigma} q_s$ given by natural transformation $p^* \Rightarrow \int_{s:\sigma} q_s^*$.

$$\mathrm{pt}(\mathcal{E})_{\sigma}(p, q_s) = \mathrm{Nat}_{E \in \mathcal{E}}(E_p, \int_{s:\sigma} E_{q_s})$$

this is a categorification of

$$p \preceq_{\sigma} q_s \Leftrightarrow (\forall U \text{ open}) \ p \in U \Rightarrow (\forall s : \sigma) \ q_s \in U_s.$$

Reconstruction theorem [Saa25, Ham25, vGMT26]

Let $\mathcal{E}$ be a topos with enough points, $\mathcal{E} \simeq \mathrm{vuCat}(\mathrm{pt}(\mathcal{E}), \mathrm{Set})$.

Remark that this construction also works for the models of any small sketch.

Removing the ultra

Removing the ultra: why?

We categorified convergence of ultrafilter to extend Makkai's ultracategories, themselves built to fit model of first-order theories. However, **now that we work with virtual ultraproducts there is no reason to restrict ourselves to ultrafilters.**

Vu-categories categorified the convergence of ultrafamilies, but the convergence of filters is more general and constructive.

We can define a filter analogue to vu-categories, called **vf-categories**. The definition is word for word the same, removing the ultra when needed.

We can then define the vf-category of points of a topos and state an analogue reconstruction theorem.

Theorem (Saadia–Tarantino, **constructive proof in progress!)**

Let $\mathcal{E}$ be a topos with enough points, $\mathcal{E} \simeq \text{vfCat}(\text{pt}(\mathcal{E}), \text{Set})$.

Recovering vu from vf?

There is a forgetful functor $U : \mathbf{vfCat} \rightarrow \mathbf{vuCat}$, but defining a functor in the other direction is not easy as the reindexing induces some vf-arrows from the existence of vu-arrows.

We will introduce **tautness**, an algebraic condition prescribing the behavior of the reindexings of arrows, and we will show that $U : \mathbf{vfCat} \rightarrow \mathbf{vuCat}$ restricts to an equivalence

$$\mathbf{vfCat}_{\mathbf{taut}} \simeq \mathbf{vuCat}_{\mathbf{taut}}.$$

Thus the ‘combinatorics of ultrafilters’ does not play any role in taut vu-categories.

Note that in particular taut ultracategories can be seen as taut vf-categories, suggesting that one can work constructively with ultracategories.

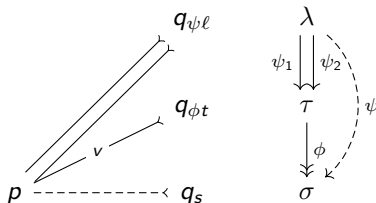
Tautness for vu-categories

Tautness: definition

In the definition of vu -categories, the behavior of the reindexing remains quite unspecified. Tautness remedies to this issue.

Reducible arrows

A vu -arrow $v : p \multimap_{\tau} q_{\phi t}$ is said **reducible along ϕ** if for each pair of arrows $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$ equalizing ϕ , we have $\psi_1^* v = \psi_2^* v$.



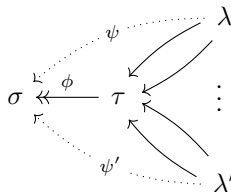
Tautness

A vu -category $\mathcal{X}$ is **taut** if each $v : p \multimap_{\tau} q_{\phi t}$ reducible along ϕ can be uniquely unthickened, i.e., there is a unique $u : p \multimap_{\sigma} q_s$ such that $v = \phi^* u$.

Tautness on the reindexing functors

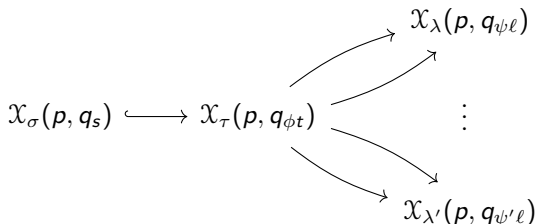
Tautness can be seen as an exactness property on the reindexing functors.

The following colimit in $\mathbb{U}$



is sent by $\mathcal{X}_{\sigma}^{p,q_s} : (\mathbb{U}/\sigma)^{\text{op}} \rightarrow \text{Set}$ to

to a limit in Set



Tautness for vu-posets

In a vu-poset any vu-arrow $p \preceq_{\tau} q_{\phi t}$ is reducible along ϕ , so tautness expresses that a vu-inequality $p \preceq_{\tau} q_{\phi t}$ can always be unthickened $p \preceq_{\sigma} q_s$.

Proposition

A small vu-poset is taut if and only if it is the points of a topological space.

We have a reflection

$$\text{TopSp} \simeq \text{vuPos}_{\text{small}, \text{taut}} \begin{array}{c} \xleftarrow{\quad t \quad} \\ \perp \\ \xrightarrow{\quad} \end{array} \text{vuPos}_{\text{small}}$$

Any small vu-poset X can be **tautified** into $t(X)$

$$p \preceq_{\sigma}^{t(X)} q_s \quad \text{if} \quad p \preceq_{\sigma}^X q_{\phi t} \text{ for some } \phi : \tau \twoheadrightarrow \sigma$$

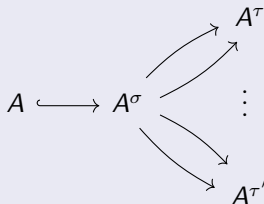
In general, $X \hookrightarrow t(X)$ is a *not full* sub-vu-poset of $t(X)$.

Tautness for ultracategories

For $A \in \mathcal{M}$ a taut ultracategory, $\mathcal{M}(-, A^\sigma) = \mathcal{M}_\sigma(-, A)$. Hence by Yoneda,

Proposition

The diagonal $A \hookrightarrow A^\sigma$ is a monomorphism, and more precisely is the join equalizer of all the pairs $A^\sigma \rightrightarrows A^\tau$ for $\tau \rightrightarrows \sigma$.



In an ultracategory, no axiom ensures that the diagonal $A \rightarrow A^\sigma$ is monic, **but in a taut ultracategory it is (and it is the limit of all the $A^\sigma \rightrightarrows A^\tau$)**.

Tautness for ultracategories

Actually we have the more general statement.

An ultracategory $\mathcal{M}$ is taut if and only if for any $\tau \twoheadrightarrow \sigma$ and $(A_s)_{s:\sigma}$, the following diagram is an equalizer

$$\begin{array}{ccc} \int_{s:\sigma} A_s & \xrightarrow{\phi^*} & \int_{t:\tau} A_{\phi t} \\ & & \swarrow \quad \searrow \\ & & \int_{\ell:\lambda} A_{\psi \ell} \\ & & \vdots \\ & & \int_{\ell:\lambda'} A_{\psi' \ell} \end{array}$$

Proposition

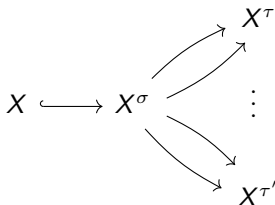
Set is taut.

Corollary

$\text{pt}(\mathcal{E})$ is taut for $\mathcal{E}$ a topos.

Tautness as internal completeness

We study tautness for the ultracategory of metric spaces. Let X be a metric space.



The image of $X \hookrightarrow X^\sigma$ is given by the sequences $(x_s)_{s:\sigma}$ that converges in X .

A σ -family $(x_s)_{s:\sigma}$ is equalized by all pairs of maps $\tau \rightrightarrows \sigma$ if and only if for any $\epsilon > 0$ there exists a closed C_ϵ of diameter $< \epsilon$ such that

$$(\forall s : \sigma) \ x_s \in C_\epsilon,$$

hence if and only if it is Cauchy.

Then the above is an equalizer precisely if X is Cauchy complete.

The ultracategory of metric spaces is not taut, **but we conjecture that its tautification is the ultracategory of complete metric spaces.**

Tautness as a right lifting property

Tautness is well justified for vu-posets. The following property shows that one can deduce the tautness only from the vu-poset case.

Proposition

A vu-category $\mathcal{X}$ is taut if and only if it is right orthogonal to all $X \rightarrow t(X)$ for X a small vu-poset,

$$\begin{array}{ccc} X & \longrightarrow & \mathcal{X} \\ \downarrow & \nearrow \text{dashed} & \\ t(X) & & \end{array}$$

i.e., if and only if

$$\text{vuCat}(t(X), \mathcal{X}) \xrightarrow{\sim} \text{vuCat}(X, \mathcal{X})$$

is an equivalence of categories.

Proof idea

We fix an ultrafilter σ and consider X the (non full) sub-vu-poset of $A_{1,\star}$, given by the $a \preceq_\tau b$ such that there exists a $\tau \rightarrow \sigma$.

In X there is a vu-arrow $a \preceq_\sigma b$ reducible along $\sigma \twoheadrightarrow \star$ but that cannot be reduced, so X is not taut and the tautification of $t(X)$ is $A_{1,\star}$.

$$\begin{array}{ccc} X & \xrightarrow{\quad} & \mathcal{X} \\ \downarrow & \nearrow \text{dashed} & \\ t(X) = A_{1,\star} & & \end{array}$$

A map $X \rightarrow \mathcal{X}$ corresponds exactly to a vu-arrow $p \preceq_\sigma q$ reducible along $\sigma \twoheadrightarrow \star$, and a map $t(X) \rightarrow \mathcal{X}$ corresponds exactly to a vu-arrow $p \preceq_\star q$.

Hence the right lifting property against $X \rightarrow t(X)$, says precisely that any vu-arrow $p \preceq_\sigma q$ reducible along $\sigma \twoheadrightarrow \star$ can be unthicken uniquely.

...the proposition can be proven by variations on the above idea.

Tautness and descent

For (S, σ) non-principal ultrafilter, $f : A_{S, \sigma} \twoheadrightarrow A_{1, \star}$ is an open surjection of topological spaces that is *not* effective descent in vuCat : when quotienting $A_{S, \sigma}$ there is no way to create the $\star$ -arrow of $A_{1, \star}$.

More precisely, the comparison map is given by $X \hookrightarrow A_{1, \star}$ the (non full) sub-vu-poset of Sierpinski of the $a \preceq_{\tau} b$ such that there exists a $\tau \rightarrow \sigma$. And $X \hookrightarrow A_{1, \star}$ is precisely the tautification of X .

But f is effective descent in $\text{vuCat}_{\text{taut}}$: when quotienting by the groupoid kernel, the tautness assumption makes the $\star$ -arrow appears.

Open surjections are not necessary effective descent in the 2-category vuCat , **but we conjecture that they are in $\text{vuCat}_{\text{taut}}$.**

Tautness vu de profil

Open vu-functor

A vu-functor $F : \mathcal{X} \rightarrow \mathcal{Y}$ is *open* if each vu-arrow $u : F(a) \multimap_{\sigma} y_s$ can be lifted up to thickening, i.e., there is a $v : a \multimap_{\tau} b_{\phi t}$ such that $F(v) = \phi^* u$.

Conjecture

Open surjections of vu-categories are effective descent in $\text{vuCat}_{\text{taut}}$.

We actually conjecture more

Profile

Following [DL22a] we call **profile** a prestack over topological spaces.

Conjecture

The 2-category vuCat embeds into profiles and a vu-category is taut if and only if it is a stack for open surjections.

Comparing VU and VF

Reindexing in vf-categories

The main difference when working with vf-categories appears on the reindexing operation. We don't only have thickening but also *specialization*.

Reindexing of vf-arrows

The reindexing $\phi^* u$ of u is called a **thickening** if ϕ is epic, and a **specialization** if ϕ is monic.

Any $\phi : \tau \rightarrow \sigma$ can be decomposed as $\tau \twoheadrightarrow \phi_* \tau \hookrightarrow \sigma$ so that any reindexing can be decomposed as a thickening of a specialization.

Two examples of specializations

The specialization of $(a_s)_{s:S} \in A^S$ along $\sigma \hookrightarrow S$ is $(a_s)_{s:\sigma}$.

The specialization of $(a_s)_{s:\sigma \sqcup \sigma'} \in A^{\sigma \sqcup \sigma'}$ along $\sigma \hookrightarrow \sigma \sqcup \sigma'$ is $(a_s)_{s:\sigma}$.

Specialization = Merging⁻¹

(Surprisingly?) the behaviour of the specialization is already specified.

Proposition

The specializations along the inclusion $\sigma_i \hookrightarrow I \cdot \sigma_i$ give a bijection

$$\mathcal{X}_{I \cdot \sigma_i}(p, q_{i,s}) \xrightarrow{\sim} \prod_{i \in I} \mathcal{X}_{\sigma_i}(p, q_{i,s})$$

whose inverse is given by merging along I .

$$\text{In particular, } \mathcal{X}_I(p, q_i) = \prod_{i \in I} \mathcal{X}_*(p, q_i) \quad \text{and} \quad \mathcal{X}_0(p,) = 1.$$

Equivalently, the reindexing functors

$$\begin{aligned} \mathcal{X}_\sigma^{p,q_s} : (\mathbb{F}/\sigma)^{\text{op}} &\longrightarrow \text{Set} \\ (\phi : \tau \rightarrow \sigma) &\mapsto \mathcal{X}_\tau(p, q_{\phi t}) \end{aligned}$$

send small coproducts to small products.

Tautness for vf-categories

As $\mathbb{F}$ is regular, tautness is easier to express for vf-categories.

Tautness for vf-categories

A vf-category $\mathcal{X}$ is taut if for each cover $\phi : \tau \twoheadrightarrow \sigma$ in $\mathbb{F}$ the following is an equalizer.

$$\mathcal{X}_\sigma(p, q_s) \longrightarrow \mathcal{X}_\tau(p, q_{\phi t}) \rightrightarrows \mathcal{X}_{\tau \times_\sigma \tau}(p, q_s)$$

Equivalently, a vf-category is taut if the reindexing functors send regular coequalizers to equalizers.

Set is a taut vf-category (because $A^- = \mathbb{F}(-, A)$ preserves colimits) and thus any vf-category arising as the point of a topos is taut.

From vu to vf

We say that $v : p \multimap_{\tau} q_{\phi t}$ is **vf-reducible along ϕ** if $\pi_1^* v = \pi_2^* v$, for $\pi_1, \pi_2 : \tau \times_{\sigma} \tau \rightrightarrows \tau$ the two projections. A vf-category is taut if any vf-reducible arrow can be uniquely unthickened.

Proposition

For $u : p \multimap_{\tau} q_{\phi t}$ a vf-arrow of a taut vf-category. If u is vu-reducible along ϕ , then u is also vf-reducible along ϕ .

Corollary

$U : \mathbf{vfCat} \rightarrow \mathbf{vuCat}$ restricts to $U : \mathbf{vfCat}_{\text{taut}} \rightarrow \mathbf{vuCat}_{\text{taut}}$.

Our goal is to show that $U : \mathbf{vfCat}_{\text{taut}} \rightarrow \mathbf{vuCat}_{\text{taut}}$ is an equivalence.

The vf-arrows of a vu-category

We start from a vu-category $\mathcal{X}$ and we want to define a vf-structure on $\mathcal{X}$.

Let's first note the following property of $A_{S,\sigma}$.

Proposition

Let p and $(q_s)_{s \in S}$ be points of $\mathcal{X}$ a vf-category.

Then vf-arrows $p \multimap_{\sigma} q_s$ can be identified with vf-functors $F: A_{S,\sigma} \rightarrow \mathcal{X}$ such that $F(a) = p$ and $F(b_s) = q_s$ for all $s \in S$.

We thus call $A_{S,\sigma}$ the **walking (S, σ) -arrow**. This suggests the following definition.

Vf-arrows of a vu-category

For p and $(q_s)_{s \in S}$ points of a vu-category $\mathcal{X}$ and σ a filter on S .

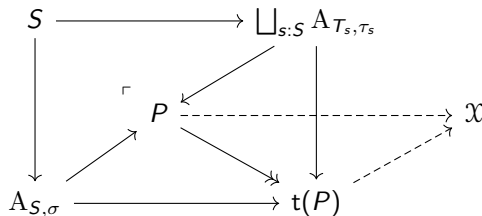
A σ -arrow $p \multimap_{\sigma} q_s$ is given by a vu-functor $F: A_{S,\sigma} \rightarrow \mathcal{X}$ such that $F(a) = p$ and $F(b_s) = q_s$ for $s \in S$.

It is easy to define the identities and the reindexings. However the composition is a bit more tricky.

Composing vf-arrows of a vu-category

We give ourselves vu-functors $F : A_{S,\sigma} \rightarrow \mathcal{X}$ and $G : A_{T,\tau_s} \rightarrow \mathcal{X}$ with $F(b_s) = G_s(a)$ and we want to define a composite $A_{\bigsqcup_{s \in S} T_s, \sigma \cdot \tau_s} \rightarrow \mathcal{X}$.

We get a map out of the following pushout P computed in vu-categories.



P is a vu-poset, but in general P is not taut, however the tautification of P contains $A_{\bigsqcup_{s \in S} T_s, \sigma \cdot \tau_s}$. Hence using the tautness of the vu-category $\mathcal{X}$, we can lift $P \rightarrow \mathcal{X}$ to a vu-functor $t(P) \rightarrow \mathcal{X}$ that restricts to a vu-functor $A_{\bigsqcup_{s \in S} T_s, \sigma \cdot \tau_s} \rightarrow \mathcal{X}$.

Recovering a vf-category from its restricted vu-category

We get a vf-category that we denote $F(\mathcal{X})$, this vf-category is taut and extends $\mathcal{X}$.

$$\begin{array}{ccc} & \overset{F}{\curvearrowright} & \\ & \perp & \\ \text{vuCat}_{\text{taut}} & \xleftarrow{U} & \text{vfCat}_{\text{taut}} \end{array}$$

It only remains to show $\mathcal{X} \cong F U \mathcal{X}$, i.e., that starting from a taut vf-category $\mathcal{X}$, the above construction applied to $\mathcal{X}$ seen as a vu-category recovers $\mathcal{X}$.

This amounts to the following result.

Lemma

Let p and $(q_s)_{s \in S}$ points of a *taut* vf-category $\mathcal{X}$.

Then vf-arrows $p \multimap_{\sigma} q_s$ are in bijective correspondence with vu-functors $F : A_{S,\sigma} \rightarrow U(\mathcal{X})$ with $F(a) = p$ and $F(b_s) = q_s$.

Lemma

Let p and $(q_s)_{s \in S}$ points of a *taut* vf-category $\mathcal{X}$. Vf-arrows $p \multimap_{\sigma} q_s$ are in bijective correspondance with vu-functors $F : A_{S,\sigma} \rightarrow U(\mathcal{X})$ with $F(a) = p$ and $F(b_s) = q_s$.

Proof

From a vf-arrow $p \multimap_{\sigma} q_s$, we have a functor $A_{S,\sigma} \rightarrow \mathcal{X}$, and by applying U a functor $A_{S,\sigma} \rightarrow U(\mathcal{X})$.

Conversely we take such a vu-functor $F : A_{S,\sigma} \rightarrow U(\mathcal{X})$.

We denote by $L := \{\sigma_{\ell} \hookrightarrow \sigma \mid \sigma_{\ell} \in \mathbb{U}\}$ the set of ultrafilter extending σ .

For each $\ell \in L$, we can consider $F(a \preccurlyeq_{\sigma_{\ell}} b_s) : p \multimap_{\sigma_{\ell}} q_s$. Merging them all together we get a vf-arrow

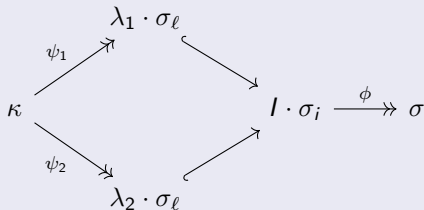
$$F(a \preccurlyeq_{\sigma_{\ell}} b_s) \circ_L d_L^p : p \multimap_{L \cdot \sigma_{\ell}} q_s.$$

We claim that this vf-arrow is reducible along the cover $\phi : L \cdot \sigma_{\ell} \twoheadrightarrow \sigma$, giving a vf-arrow $p \multimap_{\sigma} q_s$ as wanted.

proof of the claim

It remains to show that $F(a \preceq_{\sigma_\ell} b_s) \circ_L d_L^p$ is vu-reducible along $\phi : L \cdot \sigma_\ell \twoheadrightarrow \sigma$.

Notice that any map $\kappa \rightarrow I \cdot \sigma_i$ for κ ultrafilter factorizes as $\kappa \twoheadrightarrow \lambda \cdot \sigma_\ell \hookrightarrow I \cdot \sigma_i$ for some ultrafilter λ on I .



We want to show $\psi_1^*(F(a \preceq_{\sigma_\ell} b_s) \circ_{\lambda_1} d_{\lambda_1}^p) \stackrel{?}{=} \psi_2^*(F(a \preceq_{\sigma_\ell} b_s) \circ_{\lambda_2} d_{\lambda_2}^p)$

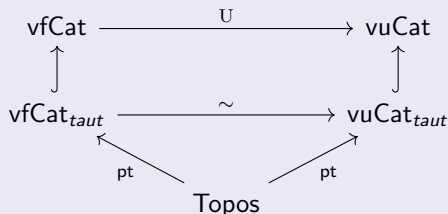
that we can rewrite $\psi_1^* F(a \preceq_{\lambda_1 \cdot \sigma_\ell} b_s) \stackrel{?}{=} \psi_2^* F(a \preceq_{\lambda_2 \cdot \sigma_\ell} b_s)$

and both side are equal to $F(a \preceq_\kappa b_s)$.

Conclusion

The full picture

Theorem: $\text{vu} = \text{vf}$



Together with the constructive proof of

Theorem: vf -reconstruction

Let $\mathcal{E}$ be a topos with enough points, $\mathcal{E} \simeq \text{vfCat}(\text{pt}(\mathcal{E}), \text{Set})$.

we recover the usual reconstruction theorem for toposes with enough points from their vu -categories of points. Thus isolating the use of the ultrafilter principle.

Escaping filters?

We propose to call the notion $\text{vuCat}_{\text{taut}} \simeq \text{vfCat}_{\text{taut}}$ (together with a size/accessibility condition) **convergence 1-space**.

We freed ourselves from the ultrafilter shape, can we go further and erase the filter shape?

We conjectured that convergence 1-spaces embed into profiles, can we characterize the image?

$$\text{TopGrpd} \dashrightarrow \text{vuCat}_{\text{taut}} \simeq \text{vfCat}_{\text{taut}} \hookrightarrow \text{Profile}$$

On the other side, we conjecture that every convergence 1-space can be represented as a topological groupoid, can we see them as some localization?

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